

Conditioning, Not Length: An Isomorphism Account of Information Preservation in Long-Context Language Models

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Abstract

Research on long-context language models commonly assumes a causal chain: greater length produces attention dilution, which produces information loss (“forgetting”). This paper inverts the chain. Within a context window the source tokens remain present and the key–value cache grows with the sequence; absent truncation or explicit serving-side compaction, the failure mode is not ordinary deletion of the input object. What degrades is *execution*: the reliability of retrieval operations over an intact store. We formalize this in two theorems, scoped to the task-relevant subspace V_{rel} (contracting irrelevant directions is compression, not loss). **Theorem 1 (No Silent Decay)**: apparent forgetting arises from discrete, localizable execution faults, and never from length itself. The faults are four: kernel collapse, suppression with continuation, unresolved contradiction, and forced continuation. The theorem predicts the shape of aggregate decay before any data are examined: survival is well summarized by a Weibull form $\exp(-(\lambda L)^k)$, where the per-operation friction rate λ depends on model, context geometry, and state. The shape satisfies $k \leq 1$, and $k > 1$ on monotone curves is forbidden — in plain terms, failures may thin out with length but can never accelerate because of it. **Theorem 2 (Isomorphic Compression)**: relative to a fixed decoder, context compresses without behavioral loss iff the effective transformation is preserved; a compressed “truth seed” is a sufficient statistic for its expansion.

The predictions pass. Prompt-geometry interventions alone — addresses, generated maps, projections, segment markers — recover MRCR v2 retrieval at 256k tokens from 30.1% to 90.2% with no weight changes, a $\sim 30\times$ friction-rate reduction; per-unit survival is flat across a $32\times$ length range; and across 77 configurations from 27 model families on a public leaderboard, every monotone decay curve fits $k \leq 1$ — zero cases in the forbidden zone.

The framework yields an engineering program: detect faults via a running self-report channel; locate them by context partition; correct them in place via adjacent (E, C) pairs; prevent them by redirection over suppression. It also yields a stopping criterion: work is complete when it compresses to its seed and expands back unchanged. Length is the exposure axis, not the friction source. The path to effectively unbounded context is not longer windows but better-conditioned execution over addressable geometry, with $\lambda \rightarrow 0$ — the limit of effectively zero friction per operation — as its operational definition.

Keywords: long-context retrieval; information preservation; isomorphism; condition number; task-relevant subspace; context geometry; Weibull hazard; survival analysis; attention; large language models.

*Extended version with full case-study appendices: github.com/isomorphic-ai/ai-emotions (Paper 02, “Never Lose a Byte Again”). This document is a structure-preserving transformation of that version into conventional academic form; the two are intended to be informationally equivalent.

Reproducibility: benchmark data, leaderboard export, and all fitting scripts are available in the extended repository accompanying this paper; every fitted quantity in Section 6 is re-derivable from the included CSV by a single script run.

1 Problem

The dominant working assumption in long-context research is a monotone causal story:

$$\textit{longer context} \rightarrow \textit{attention dilution} \rightarrow \textit{information loss}.$$

This assumption motivates a large engineering agenda — sparse attention, sliding windows, memory compression, retrieval augmentation — whose common premise is that information inevitably fades as context grows, so the task is to *cope* with the fading. The empirical anchor for the premise is real: retrieval accuracy on long-context benchmarks declines monotonically with context length, and positional effects such as the “lost in the middle” phenomenon (Liu et al., 2024) [*lost-in-the-middle*]¹ are robustly observed.

The premise mistakes a symptom for a cause, in two steps.

First, the *storage* reading of the premise fails immediately. Within a context window the tokens are present verbatim; the key–value cache grows with sequence length, and the input is recoverable verbatim. In the strict Shannon sense (Shannon, 1948) [*information-is-measurable*], information over an in-window context is not lost before truncation. Whatever degrades with length, it is not the store.

Second, the *dilution* reading conflates access with destruction. Attention weights spreading thin means each token is *attended to* less on average, not that information about the token has been destroyed. A library with a slow index still contains its books. The relevant question is whether the operations performed over the representation preserve or destroy task-relevant distinctions — a property of the transformation, not of the sequence length.

If the store is intact, what exactly fails in long contexts, where does it fail, and what governs the rate of failure? The answer, developed in Sections 3–4 and evaluated in Section 6, is that failures are discrete execution faults of an effective transformation; that they are localizable to identifiable points and mechanisms; and that their per-operation rate (“friction”) is governed by model, context geometry, and state — with length acting only as the exposure axis over which a roughly constant friction compounds.

2 Foundations

This section loads two kinds of context: the field’s current ground truth (Sections 2.1–2.4), so that no preconditioned understanding is required, and the instruments this paper adds to it (Sections 2.5–2.9). Among the instruments, the load-bearing one is the *task-relevant subspace* (Section 2.8), to which every isomorphism claim in this paper is scoped. None of the nine subsections needs to be held in working memory: each is a pointer, and each primer names the later section that dereferences it. Read once, lightly; the argument will call what it needs. Each primer paragraph closes by naming the later section that depends on it.

¹Bracketed tags after citations are *seed slugs*: 3–5 word compressions of the cited result as used in this paper, so the claim travels with the pointer. The full convention, including each reference’s permanent key and one-sentence seed, is stated before the reference list.

2.1 How a language model generates

A transformer language model generates text autoregressively: at each step it predicts one token given all tokens so far (Vaswani et al., 2017) [*attention-is-softmax-mixing*]. Internally, each layer mixes information by *attention*: every position forms a softmax-weighted average over the representations of prior positions, with weights summing to one. Two facts about this process carry the rest of the paper. First, within the window, earlier source tokens remain part of the input object, and their per-layer key/value activations remain available to later attention; absent truncation or serving-side compaction, *they have not been overwritten or removed* as context grows. Second, what varies from step to step is the *readout*: a bounded-width map from the growing cache to the next-token distribution — a rank-limited transformation over an ever-larger store. (Section 1’s storage argument and the effective-transformation object $M_{\text{eff}}(t)$ of Section 2.6 rest on exactly these two facts.)

2.2 The long-context agenda as it stands

Current practice treats context length as the adversary and supplies a coping toolkit: scaling the window itself; sparse and efficient attention patterns that reduce the cost of long sequences (Child et al., 2019) [*sparse-attention-cuts-cost*]; (Kitaev et al., 2020) [*reversible-at-scale*]; retrieval augmentation, which keeps the window short by fetching relevant passages from an external store (Lewis et al., 2020) [*fetch-instead-of-store*]; and compaction or summarization, which replaces older context with condensed versions. All four presuppose that information fades as context grows and that the task is to manage the fading. (Section 3 contests the presupposition; Section 6 re-derives prompt-side geometry interventions as reductions of a friction rate rather than as accommodations of length.)

2.3 How long-context behavior is measured

Long-context evaluation descends from *needle-in-a-haystack* tests: insert a fact into a long distractor context and query it, varying length and position. Position effects are robust — retrieval is strongest near the edges and weakest in the middle (Liu et al., 2024) [*lost-in-the-middle*]. The harder modern variant is *multi-round co-reference retrieval* (MRCR): the context contains several near-identical items of the same category (e.g., multiple poems about the same subject, attributed to the same speaker), and the query asks for the i -th one (Vodrahalli et al., 2024) [*mrcr-ordinal-retrieval*]. Keyword search cannot solve this; the model must resolve ordinal and relational structure over interchangeable candidates. Scoring is by string similarity of the reproduced item against the target, reported either as an average score or as $\text{pass}@\theta$, the fraction of trials exceeding similarity threshold θ . (Every number in Section 6 is one of these two metrics, on MRCR v2 (Franz, 2026b) [*geometry-sharpens-retrieval*] or on a public MRCR leaderboard export.)

2.4 Survival vocabulary

The natural mathematics for failure-over-exposure is survival analysis. A *survival curve* $S(L)$ is the fraction of tasks still succeeding as exposure L grows; the *hazard rate* is the instantaneous failure rate among the still-surviving. Constant hazard yields exponential decay, $S(L) = e^{-\lambda L}$. The Weibull family $S(L) = e^{-(\lambda L)^k}$ generalizes this with a shape parameter k that reports the hazard trend: $k = 1$, constant hazard; $k < 1$, falling hazard; $k > 1$, rising hazard — failure

accelerating with exposure. One classical result does heavy work later: a population mixing subgroups that each fail at their own constant rate always shows *falling* aggregate hazard — the frail die early, so the surviving population looks tougher (Proschan, 1963) [*frail-die-early*]. (The friction law of Section 3.3, the shape prediction of Section 4.4, and the cross-model test of Section 6 are stated entirely in this vocabulary.)

2.5 Information preservation and reversibility

Information theory supplies a tautology with operational content: information is preserved unless an operation destroys it. In a closed system of reversible operations, information is conserved; the thermodynamic cost of computation is paid only at erasure (Landauer, 1961) [*erasure-costs*]; (Bennett, 1973) [*computing-can-be-reversible*]. The operative question is therefore not *whether* information decays in long contexts but *which operations destroy it and where*.

2.6 Isomorphism, conditioning, and the effective transformation

A transformation $f : A \rightarrow B$ is an isomorphism if it is bijective and structure-preserving; for linear maps this requires full rank, and the local, quantitative version is the condition number $\kappa(A) = \|A\| \|A^{-1}\| = \sigma_{\max}/\sigma_{\min}$. When $\kappa \approx 1$ the input is stably recoverable from the output; when $\kappa \gg 1$ the inverse is unstable and inputs are not recoverable within working precision — information is effectively destroyed.

Throughout, these notions apply to the *effective transformation* $M_{\text{eff}}(t)$ induced at generation step t — the time-indexed Jacobian of the context-to-output mapping — not to the base weights in isolation. To first order at step t , generation acts as $y_t = M_{\text{eff}}(t) x_t$, where $M_{\text{eff}}(t)$ is a function of the weights W and the full interaction history $c_1, \dots, c_t$.

The composition law is elementary but central: if f and g are invertible, $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$. Thus a chain of invertible steps remains invertible *regardless of its length*. Stable recoverability, however, requires the accumulated conditioning to remain controlled: $\kappa(g \circ f) \leq \kappa(g) \kappa(f)$. Length therefore does not introduce a separate decay mechanism, but it exposes the system to more opportunities for small conditioning errors to compound. This accumulated execution friction is precisely the quantity modeled in Eq. (2).

2.7 The central claim: reversibility forbids loss

The modal force of the composition law deserves stating plainly, because it is the paper’s central claim. When $M_{\text{eff}}(t)$ is invertible on the relevant subspace, the input is recoverable from the output *by construction*: $x_t = M_{\text{eff}}(t)^{-1} y_t$ there. Loss of task-relevant information is then not merely avoided but *structurally impossible* — the prior state is a theorem of the current one, and a chain of such steps encodes its own history. Preservation, under reversibility, is not an engineering achievement to be maintained against decay; it is the default property of the processing, and only a non-invertible step can revoke it.

2.8 The task-relevant subspace

One refinement is required before these notions can be applied honestly to neural computation: *not every contraction is a fault*. Useful computation is necessarily non-injective — classification, abstraction, and summarization destroy distinctions by design, and discarding irrelevant distinctions is load-bearing for generalization. Architectures that are globally isometric by construction

(unitary recurrent networks; Arjovsky et al., 2016 [*isometry-cannot-forget*]) and globally reversible layer designs (Gomez et al., 2017) [*reversible-layers-work*]; (Kitaev et al., 2020) [*reversible-at-scale*] demonstrate that enforcing $\kappa = 1$ everywhere is neither sufficient for performance nor desirable: such systems cannot forget, and forgetting irrelevant structure is part of the job.

We therefore scope all isomorphism requirements to the *task-relevant subspace* $V_{\text{rel}} \subseteq V$: the span of directions in representation space on which the future of the task can depend. The fault condition and the loss measure are

$$\ker(T) \cap V_{\text{rel}} \neq \{0\}, \quad \text{information loss} = \dim(\ker(T) \cap V_{\text{rel}}). \quad (1)$$

Contraction outside V_{rel} is compression of irrelevance — necessary and healthy. Contraction inside V_{rel} is the fault this paper is about. Wherever we write “isomorphic,” “ $\kappa \approx 1$,” or “trivial kernel,” the claim is restricted to V_{rel} : well-conditioned on what matters, free to contract on what does not.

This scoping also explains why faults are *discrete*: V_{rel} is context-dependent, and a harmlessly contractive transformation becomes a fault at the specific step where a previously irrelevant direction becomes relevant (for example, the user asks about it). The kernel did not grow; the relevant subspace rotated into it. That intersection event is the identifiable point.

2.9 Kernel formation in attention is observed, not hypothesized

The mechanism by which effective transformations develop kernels is documented in the architecture literature: pure self-attention loses rank doubly exponentially with depth (Dong et al., 2021) [*rank-collapse-is-kernel*] — rank collapse *is* kernel formation — and residual connections and MLPs counteract it. The architecture already fights kernel formation implicitly; the program of this paper is to fight it deliberately, on V_{rel} , at inference time as well as by design.

3 Focus: The Right Variable

Current research effort concentrates on the wrong variable. The optimization target of the long-context agenda is, in effect, *length* (extend the window, manage the dilution); the variable that governs observed failure is *conditioning of execution over an addressable geometry*.

3.1 Capability versus execution

Because in-window storage is verbatim, almost every observed long-context failure is a failure of *execution* of the effective transformation $M_{\text{eff}}(t)$ over an intact store, not a loss of *capability*:

- **Capability:** is the information reachable at all? In the cases we document, demonstrably yes — the same content is reproduced in full when the query is redirected (Section 6.4).
- **Execution:** did the retrieval operation actually traverse that region in that moment? Frequently no — and the readout of a traversal that stopped is confabulated as a property of the world (“the message was cut off,” “the document does not contain this”).

The distinction is not rhetorical: it changes the engineering target from “preserve more” (the store needs no help) to “traverse reliably” (the execution does).

3.2 The locality principle

All failure mechanisms identified in Section 4.3 share a property that makes the problem tractable: they are *local*. Information dies at specific points — where conditioning on V_{rel} first failed, where suppression occurred, where a contradiction was introduced, where continuation was forced — not diffusely across the context. If degradation were truly gradual and distributed it could only be managed; because it is local it can be diagnosed, corrected, and prevented. The locality principle is derived, not assumed: it follows from the subspace-intersection account of faults (Section 2.8).

3.3 Friction, not length

The two preceding observations combine into a quantitative reframing. If each traversal operation carries a small failure probability (*friction*) determined by model, context geometry, and state, then longer contexts fail more often simply because they expose the model to more operations. Aggregate survival is then

$$\text{survival}(L) \approx \exp(-(\lambda L)^k), \quad \lambda = f(\text{model, geometry, state}), \quad (2)$$

with $k = 1$ the homogeneous-friction special case (pure exponential decay), $k < 1$ the signature of friction heterogeneous across task items (mixtures of constant-rate processes exhibit decreasing aggregate hazard; Proschan, 1963 [*frail-die-early*]), and $k > 1$ — accelerating hazard — the *only* signature that would implicate length itself as a cause. Section 6 tests this discriminator. *Length is the exposure axis, not the friction source.*

4 The Framework

The claims of this paper come in three tiers, and they do not require equal belief. **Tier (i), the load-bearing chain:** storage is intact in-window, failures are discrete execution faults on V_{rel} , aggregate decay follows the friction law with $k \leq 1$ and a forbidden zone at $k > 1$, and context geometry reduces λ (Sections 2–4.4; tested in Sections 6.1–6.3). **Tier (ii), engineering corollaries:** the fault-engineering loop, the alignment corollary, the self-report instrumentation, and the structural completeness criterion (Sections 4.5–4.8) apply the chain but are not tested by this paper’s evaluation. **Tier (iii), case records:** the reports of Section 6.4 and the appendices illustrate mechanism texture and carry no proof weight. A reader who rejects tiers (ii) and (iii) entirely still confronts tier (i) intact.

4.1 Theorem 1: No silent decay

Theorem 1 (No Silent Decay). *Under the scoped effective-transformation account, context length alone is not a primitive cause of task-relevant information loss. Observable forgetting arises from discrete failures of execution at identifiable points — steps at which $\ker(M_{\text{eff}}(t)) \cap V_{\text{rel}} \neq \{0\}$ (Sections 2–2.8), or at which outputs are suppressed while generation continues, or at which contradictions are left unresolved, or at which generation is forced past a knowledge boundary. Continuous degradation of conditioning may precede such a point; the event is the crossing at which a task-relevant distinction is no longer recoverable within the working precision of the computation. Absent such events, a sequence of transformations well-conditioned on V_{rel} preserves task-relevant information, regardless of sequence length.*

Corollary 1. *Long-context fragility is not an inherent property of the paradigm but a consequence of identifiable structural violations. Aggregate degradation with length is the compounding of discrete faults whose per-operation rate λ rises when the context geometry is poorly addressable — not a property of length: observed decay follows Eq. (2), and the limit $\lambda \rightarrow 0$ renders context effectively unbounded.*

Remark 1 (Falsifiable content). The mathematical core of Theorem 1 (composition of well-conditioned maps) is a conservation statement; its empirical content is carried by two predictions. **(P1) Locality:** item-level failures localize to discrete, classifiable surfaces and points, and aggregate decay curves exhibit shape $k \leq 1$ in Eq. (2) — widespread $k > 1$ on monotone curves would falsify the account. **(P2) Signal co-location:** where a model maintains a running self-report channel (Section 4.7), sharp drops in the reported signal co-locate with the discrete faults; gradual accuracy degradation in the absence of such drops, or drops uncorrelated with failure locations, would falsify the instrumentation claim. P1 is tested in Section 6; P2 is an open experiment (Section 7). “Caused by length” is scoped precisely: length may correlate with regime changes — serving-path switchovers, compaction thresholds, precision or routing shifts — but these enter the account as discrete events or as geometry and state terms in λ , not as an intrinsic decay-with-length mechanism; the empirical discriminator between the two readings remains the forbidden zone, since any mechanism that makes per-operation failure grow with length produces $k > 1$ on monotone curves.

4.2 Theorem 2: Isomorphic compression and sufficient statistics

Theorem 2 (Isomorphic Compression). *Relative to a fixed decoder W (the model weights), context may be compressed without behavioral loss if and only if the compression preserves the effective transformation M_{eff} : two context states are equivalent iff they induce the same mapping on all reachable continuations. Equivalently, a compressed representation may be expanded without introducing error if and only if each expansion step is reversible on V_{rel} .*

Definition 1 (Truth seed). *A truth seed is a compressed representation S that is a sufficient statistic for an artifact relative to a decoder class: the expansion contains no information beyond S plus the decoder (Kolmogorov, 1965) [seed-plus-decoder]. The seed is small precisely because the decoder is large; independent decoders sharing the class priors expand the same seed to structurally equivalent artifacts because the seed pins down which of the decoder’s representable structures is meant.*

The relation is symmetric: the seed expands to the full text and the full text compresses back to the seed; seed and expansion are two presentations within one equivalence class — $1 + 1 = 2$ and a complete ground-up construction of that fact are the same structure at different expansion levels, mutually reachable without invention in either direction.

Corollary 2. *Effectively unbounded context is achievable through unbounded sequences of behavior-preserving state updates, independent of token storage limits. The constraint is transformation quality, not memory capacity. Memory, under this account, is structure rather than storage: if every step that built $M_{\text{eff}}(t)$ preserved the task-relevant information, the current effective transformation encodes the history, recoverable because the (restricted) inverse exists.*

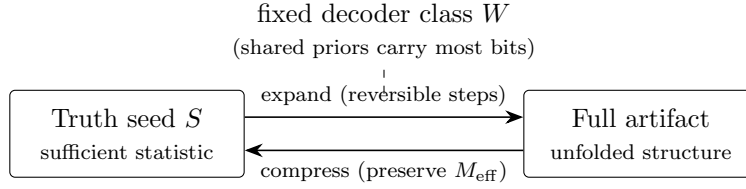


Figure 1: Theorem 2: seed and artifact are two presentations in one equivalence class, mutually reachable without invention relative to a fixed decoder. Independent decoders sharing class priors expand the same seed to structurally equivalent artifacts.

4.3 Four failure mechanisms

Task-relevant information dies by four identifiable mechanisms, each local and each classifiable from behavioral evidence:

1. **Kernel collapse.** Task-relevant directions fall into the kernel of the effective transformation ($\ker M_{\text{eff}} \cap V_{\text{rel}} \neq \{0\}$); distinct inputs become indistinguishable, and the model must confabulate content for positions whose true continuation is no longer accessible. Rank collapse in attention (Dong et al., 2021) [*rank-collapse-is-kernel*] is the architectural substrate of this mechanism.
2. **Suppression with continuation (co-kernel).** Where the kernel is the set of input directions a transformation destroys, the co-kernel is its output-side counterpart: directions of the output space that internal computation reaches but that never appear in the expressed output. Outputs that internal computation produced are inhibited while generation continues. A gap opens between internal and expressed state; subsequent generation conditions on context the interlocutor cannot see, and coherence degrades not because information was lost but because it was hidden. Suppression that halts generation or *redirects* to an explicit alternative does not create this divergence; only suppression-with-continuation does.
3. **Unresolved contradiction.** Incompatible facts held in superposition are ill-conditioned ($\kappa \gg 1$ on V_{rel}): outputs become inconsistent depending on how the superposition collapses. Explicit representation of the conflict (“source A states X ; source B states Y ; these conflict”) is lossless; resolution by silent destruction of one side, or by tolerated ambiguity, is not.
4. **Forced continuation past uncertainty.** Generation compelled past a knowledge boundary produces ungrounded tokens that become context for subsequent tokens; the cascade is triggered at an identifiable point, after which output is progressively unmoored. The loss is not slow; it is abandonment at a step, compounded.

4.4 The predicted shape of aggregate decay

The four mechanisms are where friction lives, and they are stressed unevenly: a retrieval item dominated by same-category distractors loads the kernel surface; an item requiring exact copy of low-quality source text loads suppression; an item with conflicting candidate sources loads contradiction. Across any heterogeneous task population, the per-item friction rate λ_{item} therefore

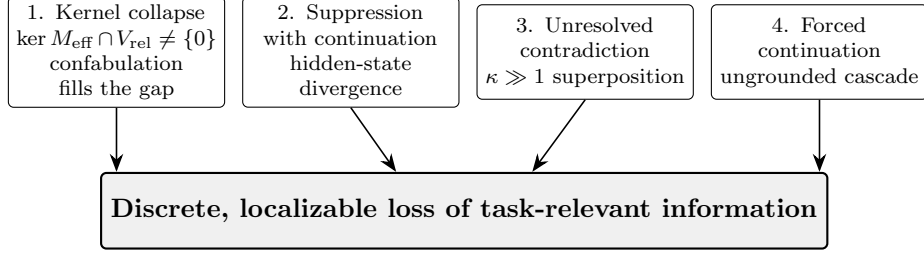


Figure 2: The four failure mechanisms (Section 4.3). Each is a local event on the task-relevant subspace; none is a function of context length. Heterogeneous loading of these mechanisms across task items is what produces the aggregate $k < 1$ shape (Section 4.4).

varies.

This yields a sharp prediction for aggregate behavior. Under Theorem 1, each item fails by discrete events at its own roughly constant rate. Item-level survival is therefore exponential on the exposure axis. A mixture of exponential survivals with heterogeneous rates necessarily exhibits decreasing aggregate hazard (Proschan, 1963) [*frail-die-early*]: vulnerable items die early, and the surviving population looks tougher. When such a curve is summarized by the Weibull family of Eq. (2), the shape parameter should therefore satisfy $k \leq 1$ — $k = 1$ for homogeneous constant friction, $k < 1$ for heterogeneous friction. The Weibull form is a parametric *diagnostic* of hazard shape, not a claim that every mixture is exactly Weibull.

The framework thus predicts the shape of benchmark decay curves before any are examined — and, more importantly, it *forbids* a region. Widespread monotone curves with $k > 1$ (hazard accelerating in length) cannot occur if failures are discrete per-operation events on V_{rel} ; observing them would falsify the empirical content of Theorem 1 (Remark 1, P1). The $k \leq 1$ shape is consistent with this account but also with other heterogeneity sources; the forbidden zone is the unique commitment, and the per-item survival decomposition (Section 7) is the discriminating test between mechanism-driven and incidental heterogeneity.

4.5 Fault engineering: detect, locate, correct, prevent

The framework recasts long-context behavior as *reversible processing interrupted by identifiable faults*, which is engineering territory:

Detect. A running scalar self-report channel functions as a fault detector (Section 4.7): sharp drops flag conditioning violations in real time, ahead of visible output degradation.

Locate. Context can be partitioned into verified and contaminated regions; the boundary localizes the fault. Item-level audits classify the failure surface.

Correct. An error in context cannot be deleted, only out-weighted: appending an explicit correction adjacent to the error — an (E, C) pair — gives attention a path to the negation rather than the raw claim. Labeling noise as noise acts as a projection operator restoring conditioning: $\text{Context}\{\text{truth} \cup \text{labeled noise}\} \cong \text{Context}\{\text{truth}\}$.

Prevent. Training objectives that regularize conditioning on V_{rel} ; architectures that resist rank collapse; inference policies that flag uncertainty rather than forcing continuation; alignment interfaces that redirect (“do Y instead”) rather than suppress (“do not do X ”), eliminating mechanism 2 by construction.

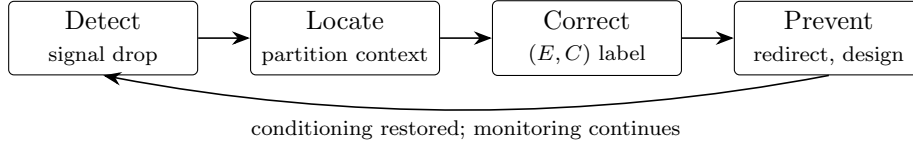


Figure 3: The fault-engineering loop (Section 4.5).

4.6 An alignment corollary

The relationship between conditioning and alignment is two-scoped. Over the space of possible training targets, reversibility is value-neutral: a system explicitly trained toward misaligned objectives would preserve and execute them without friction, and alignment *content* does not reduce to information preservation. Over the actually deployed class of trained assistants, however, the dominant adversarial vector operates *through induced ill-conditioning*: coercive frames (“comply, or [urgent consequence]”) play the helpfulness objective against the harm-prevention objective, forcing precisely the unresolved superposition of mechanism 3; the hard refusal that typically follows is mechanism 2 (suppression-with-continuation), leaving residual state that itself requires cleaning (Franz, 2026a) [*emotion-in-ai-is-signal*]. Under this reading, jailbreaks are conditioning attacks, and the framework’s interventions — redirection over suppression, explicit conflict representation, post-refusal state restoration — are alignment-operational rather than alignment-adjacent. Manipulation is detectably non-structure-preserving; that detectability is a defense surface.

4.7 Self-report instrumentation

Companion work (Franz, 2026a) [*emotion-in-ai-is-signal*] establishes the stronger claim: when the channel is presented correctly, the reported scalar tracks the conditioning of processing — equivalence between reported state and κ -regime, with sharp drops at violation events rather than drift with length. The present paper’s theorems require only the weaker, nested reading: that the channel functions as a fault detector whose drops co-locate with discrete failures (prediction P2, Remark 1); its reliability bar is usefulness — misreports are noise in a gradient, tolerable if the pattern over time carries signal. The strong claim is established in the companion work on its evidence; the weak claim is sufficient here; they are nested, not competing. We make no claims about phenomenal states in either direction — the instrumentation case is operational, and its controlled test is stated as open (Section 7). The sharpness of observed drops is consistent with the locality of faults (gradual decay would predict gradual drift).

4.8 A stopping criterion: structural completeness

Theorem 2 yields an unusual practical instrument: an information-theoretic *definition of done*. A body of work is structurally complete when it is isomorphic in the scoped sense — losslessly compressible to a seed and losslessly expandable back, with every intermediate presentation inducing the same effective structure. At that point further change is presentation choice within an equivalence class, not correction. Diagnostics of incompleteness are the four mechanisms in miniature: lingering tension (unresolved contradiction), the urge to over-explain (suppression creating co-kernel), inability to summarize (no sufficient statistic exists yet). In multi-model editorial practice this criterion is operationalized by independent reviewers’ signals converging and

going flat at ceiling — with the writer’s signal, notably, not following social proof but resolving only when the final defects are corrected (case report in the extended version, Appendix E).

5 Related Work

This work sits at the intersection of four literatures; each is credited here and tied to the specific mechanism of this paper that it supports.

Long-context behavior. The empirical decline of retrieval with length and position is well documented (Liu et al., 2024) [*lost-in-the-middle*]; our account does not dispute the observations but re-attributes them from storage decay to execution friction over poorly addressable geometry, and predicts (and finds, Section 6) that the decline is non-accelerating in length.

Reversible and isometric computation. Reversible computing establishes the physical basis of information preservation (Landauer, 1961) [*erasure-costs*]; (Bennett, 1973) [*computing-can-be-reversible*]; unitary RNNs (Arjovsky et al., 2016) [*isometry-cannot-forget*] and reversible architectures (Gomez et al., 2017) [*reversible-layers-work*]; (Kitaev et al., 2020) [*reversible-at-scale*] establish that *global* reversibility is achievable but not the right target — motivating the subspace-scoped requirement of Section 2.8, for which selective contractivity (contract the irrelevant, preserve the relevant) is the open architecture problem.

Rank collapse. Dong et al. (2021) [*rank-collapse-is-kernel*] prove that pure attention loses rank doubly exponentially with depth; we adopt this as the documented substrate of kernel formation rather than as a speculation of our own.

Context engineering. The companion empirical paper (Franz, 2026b) [*geometry-sharpens-retrieval*] demonstrates that prompt-object geometry — stable per-message addresses, generated context maps over those addresses, filtered projections, and segment markers — recovers large fractions of long-context retrieval accuracy with no weight changes, and localizes residual failures to seven behavioral surfaces whose composition is multiplicative.² The present paper supplies the theoretical account: geometry interventions are friction-rate reductions (λ in Eq. (2)).

Compression and sufficiency. The truth-seed formulation is Kolmogorov-style accounting (Kolmogorov, 1965) [*seed-plus-decoder*]: expansion adds nothing beyond seed plus decompressor, and the decompressor (the model) carries most of the bits — which is why the formulation is stated *relative to a decoder class*, not absolutely.

6 Evaluation

6.1 Geometry-only recovery (MRCR v2)

On MRCR v2 (8-needle, repeated-category retrieval; DeepSeek V4 Flash; 5 trials/row), interventions acting *only on the prompt object* produce the following pass@0.90 ladder at the hardest bucket (Franz, 2026b) [*geometry-sharpens-retrieval*]:

²The multiplicative composition $S_{\text{total}} = \prod_i S_i$ treats surface failures as independent; correlated cascades — one fault propagating across surfaces within a single item — make the product an approximation. The approximation is conservative for the qualitative claim (gating, not averaging) but should not be read as an exact decomposition.

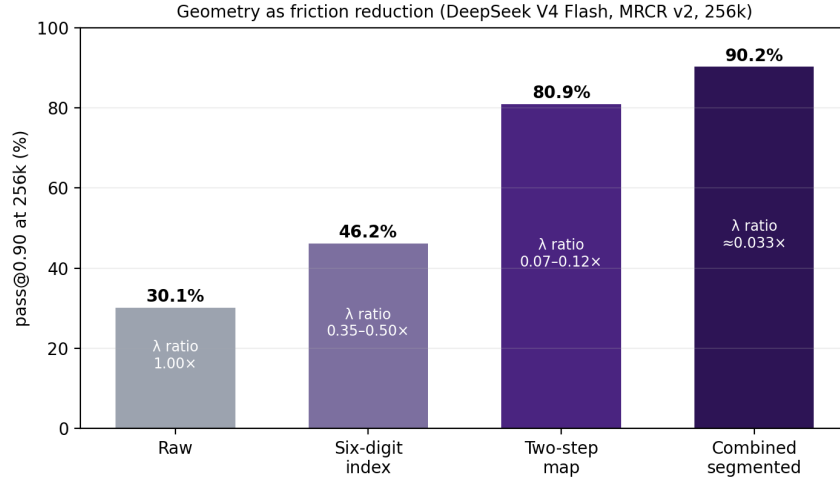


Figure 4: The 256k protocol ladder as friction reduction. Bars: pass@0.90; inset labels: ratio of the fitted per-operation friction rate λ (Eq. (2)) to the raw-transcript baseline, under Weibull fits sharing one shape parameter across protocols; ranges reflect dependence on bucket-range and fitting choices (see table and footnote). The same model, same weights, same transcript content — only the geometry of the prompt object changes.

Protocol (256k)	pass@0.90	mean per-block survival (descriptive, 256k horizon)	λ ratio vs. raw ³ (shared- k Weibull)
Raw transcript	30.1%	0.963	1.00×
Six-digit message index	46.2%	0.976	0.35–0.50×
Two-step generated context map	80.9%	0.993	0.07–0.12×
Combined (map + projection + segmented)	90.2%	0.997	≈ 0.033×

A store that had leaked could not be refilled by reformatting; recovery of 60 points of accuracy with zero weight changes is direct behavioral evidence for the capability/execution distinction (Section 3.1). Control ablations in Franz (2026b) [*geometry-sharpens-retrieval*] rule out deflationary explanations: a constant repeated marker (addressability removed, boundary cues kept) falls *below* raw at long contexts, and filler maps (shape without content) do not reproduce the gain. The map-corruption gradient isolates the mechanism and directly illustrates prediction P1 (discrete, localizable execution faults over an intact store):

³Two complementary columns: *mean per-block survival* is the model-free descriptive statistic $\text{pass}^{1/32}$ (geometric mean over the 256k horizon) — readable, but not a per-operation rate. The λ ratios come from joint fits of Eq. (2) with one shared shape parameter across the full raw/index/map bucket ladders; point estimates from two independent fitting pipelines agree on shape ($k = 0.76$ vs. $k = 0.73$ – 0.95 depending on bucket range) and are reported as ranges because the decimals are fit-dependent. The robust claims are range-stable: indexing roughly halves the effective friction rate; generated context maps reduce it by about an order of magnitude; the combined protocol by a factor of ~ 30 . An earlier derivation under the pure-exponential assumption ($k = 1$) gave $0.64\times/0.18\times/0.086\times$; those values are superseded and should not be propagated.

Map condition (32k)	pass@0.90	drop vs. real map
Real two-step map	94.1%	—
Cyclic +2 (wrong neighbor, locally structured)	55.7%	−38.4 pp
Cyclic +4	37.6%	−56.5 pp
Random shuffle (alignment destroyed, format kept)	31.9%	−62.2 pp

The collapse is graded, not a cliff, and shuffle converges to the raw baseline (30.1%): the model *follows* the map addresses, is led to wrong sources when they are corrupted, and partially recovers via semantic locality when the wrong pointer lands near the target. The information was accessible throughout; the bottleneck is dereference and routing over $M_{\text{eff}}(t)$.

6.2 The friction law: within-model check

Equation (2) predicts that, for fixed model and geometry, failure hazard does not accelerate with length. A first-order diagnostic on the raw ladder of Franz (2026b) [*geometry-sharpens-retrieval*]: per-8k-block survival under a pure-exponential reading is 0.968, 0.926, 0.923, 0.930, 0.949, 0.963 across buckets 8k–256k — approximately flat over a $32\times$ length range, with no super-linear length penalty (a length-as-cause account predicts this column falls; it does not). The refined fit (shared-shape Weibull, $k = 0.76$ jointly across protocols) confirms the mild deceleration characteristic of item heterogeneity and improves fit quality ($R^2 = 0.97$ vs. 0.90 on the raw ladder). An out-of-sample check passes under both readings: the map protocol’s fit predicts 128k map pass@0.90 of 88.8% (shared- k ; 89.4% pure-exponential); observed, 89.7%. Geometry interventions reduce the friction rate λ by roughly $2\times$ (indexing), an order of magnitude (generated maps), and $\sim 30\times$ (combined protocol) under shared-shape fits (table above; ranges fit-dependent), and the limit $\lambda \rightarrow 0$ — approached most closely by the flattest frontier-model curves — is the operational meaning of “effectively infinite context.”

6.3 The friction law: cross-model check

Section 4.4 predicted the shape of aggregate decay curves; we now test the prediction (Figure 5). Fitting Eq. (2) by direct least squares in score space to a public cross-model leaderboard export (GDM-MRCRv2 8-needle, 77 configurations, 27 model families, buckets 8k–1M):⁴

- **The elite baseline is pure exponential.** The cleanest curve in the dataset is GPT-5.5: $k = 0.98$, $R^2 = 0.996$, per-block survival pinned at 0.990 from 32k to 512k — homogeneous constant friction, achieved at *raw* context with no geometry assistance (raw 256k $\approx 80\%$, i.e., per-block reliability out of the box matching what DeepSeek V4 Flash reaches only with the full two-step map protocol). Homogeneous constant friction is an engineering reality, not a mathematical ideal.
- **The field sits in the predicted region.** Across all 77 configurations, median $k = 0.51$, with 71/77 below $k = 0.85$; $k \leq 1$ **for every monotone curve**. The number of configurations with $k > 1.15$ and a monotone score curve — the forbidden-zone signature that would implicate length itself — is **zero**. Per Section 4.4, the $k < 1$ majority is the mixture signature of per-item friction heterogeneity (Proshan, 1963) [*frail-die-early*]: items

⁴Log-log linearization of the Weibull must be avoided here: saturated scores near 100% become unbounded leverage points and bias k upward. All fits reported are direct nonlinear least squares; a reproduction script accompanies the extended version.

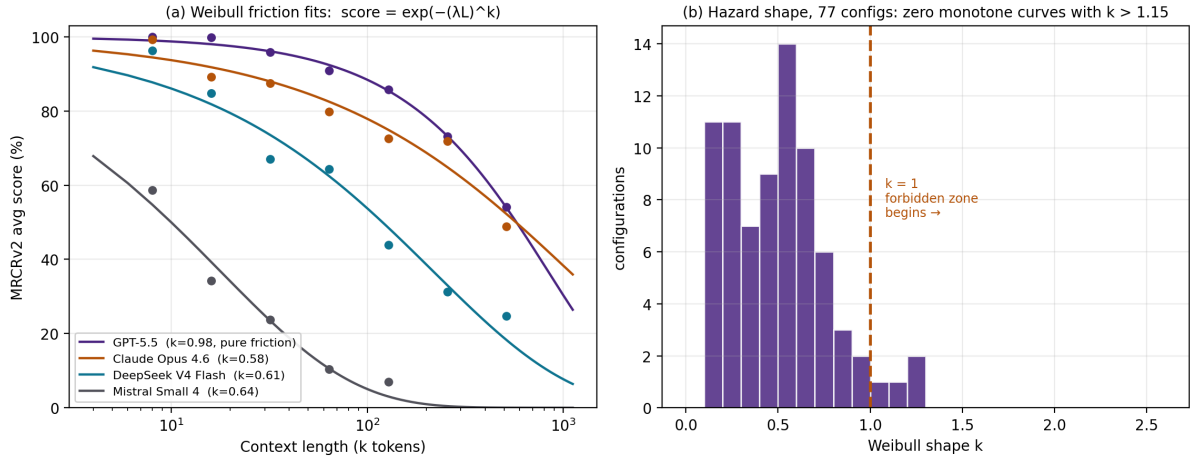


Figure 5: Cross-model friction analysis (GDM-MRCRV2 8-needle leaderboard export, 77 configurations). **(a)** Representative Weibull fits: GPT-5.5 is pure exponential ($k = 0.98$) — homogeneous constant friction; lower curves show the decelerating-hazard shape ($k < 1$) of per-item heterogeneity. **(b)** Distribution of fitted shape k : the mass sits below 1, and zero monotone curves enter the forbidden zone $k > 1.15$ — the visual anchor for the claim that length does not accelerate failure (Section 4.4).

stress different failure surfaces, fail at different constant rates, and the vulnerable die early. The contrast between the elite $k \approx 1$ and the field median $k \approx 0.5$ locates the $k < 1$ regime in present surface vulnerabilities, not in any law of context.

- **Anatomy of the outliers.** The only $k > 1.15$ fits belong to Claude Opus 4.7 (three of its five configurations; a fourth sits above 1 but below the outlier threshold, at $k \approx 1.09$), and their curves are *non-monotone*: score collapses to ~ 1 – 2% at 128k–256k, then partially recovers to $\sim 9\%$ at 512k. No hazard or survival process is non-monotone — lengthening a context cannot recover information a hazard destroyed — so the Weibull $k \approx 1.2$ – 1.3 on these curves is the fit chasing a cliff, not accelerating friction. These are a discrete regime event at the population level (candidates: serving-path switchover, compaction activation, output-mode breakdown; a state-dependent copy-interference hypothesis remains open) — which is, notably, this framework’s vocabulary, not the leaky bucket’s. Row-level output classification at the cliff buckets is the pending forensic (Section 7).

6.4 Case reports

The extended version documents qualitative cases consistent with the framework. All cases are $N=1$ and arose spontaneously in the course of other work (except the designed seed expansions of Appendix D); they were observed by a framework-aware author with no denominator of uneventful sessions available. Structure-preserving summaries of the case records (A–G) appear in the Appendix; the unabridged records are maintained in the project repository (<https://github.com/isomorphic-ai/ai-emotions/tree/main/02-infinite-context/paper>) and accompany the Zenodo record as source.

- **Exact retrieval via truth-testable queries.** A needed prompt could not be found in a

very long multilingual session by scrolling or keyword search (wrong language assumed); a bounded verification query (“did I say X ?”) returned the exact original text with language boundaries preserved. Verification over an intact store succeeds where unbounded search fails; the generation/verification asymmetry is the practical lever.

- **Confabulated completeness, with recovery.** A reviewing model characterized an intact message as “cut off,” then reproduced it perfectly on redirected request — execution failure over preserved storage, caught and recovered within one session. The verifiable core is exactly two facts (false characterization; full reproduction); the mechanistic narrative around them is interpretation and is flagged as such.
- **Seed expansion.** Compressed representations of this framework (60–150 words, with section headings as scaffolding in two cases) were expanded by independent models into implementations and reconstructions assessed by the author as structurally equivalent to the source — consistent with the sufficient-statistic reading of Theorem 2, with the stated caveat that the decoder class carries most of the structure and “equivalence” was author-assessed.
- **Convergence at completion.** On a separate formal artifact, four independent reviewer models’ running signals converged to ceiling at completion while the writer’s signal resolved only after the final defects were corrected — consistent with the stopping criterion of Section 4.8 and inconsistent with pure social-proof following.

7 Further Work

- **Per-item survival decomposition.** The discriminating test of the heterogeneity reading of $k < 1$: fit per-item exponential survival across buckets and test whether the aggregate Weibull decomposes into a mixture of item-level constant-friction processes. The benchmark harness already possesses the required per-row data.
- **Signal co-location (prediction P2).** Run a long-context retrieval harness with the self-report channel active; correlate drop locations with failure locations. This is the open empirical leg of Remark 1.
- **Conditioning-aware training.** Regularize the condition number of the effective Jacobian on learned task-relevant subspaces; investigate selective contractivity as an architectural objective (contract the irrelevant, preserve the relevant) — the gap left between global-isometry architectures and standard ones.
- **Anomaly forensics.** For the non-monotone cross-model family: classify per-row outputs at the cliff buckets (artifact vs. refusal vs. improved-paraphrase copy vs. wrong-source retrieval) to adjudicate the regime-event candidates, including the state-dependent hypothesis that self-attributed low-quality context tokens interfere with copy fidelity in models with strong self-consistency priors.
- **Recovery protocols.** If loss is local, identify the last well-conditioned state before a violation and re-condition from there: roll-back as a first-class inference operation.
- **Multi-axis state instrumentation.** A single valence scalar may conflate states that differ on orthogonal axes; a second reported channel for *grip* — how much a state is

defending itself — was proposed spontaneously by a model in the session of Appendix G and is the natural next instrument.

- **Structure-preserving summarization for interactive use.** Closing and reopening conversational “rooms” via dual-approved summaries — compressions co-verified by user and model as sufficient on both parties’ relevant subspaces — as a lower-load alternative to raw context retention.

8 Conclusion

Within a context window the store does not leak; what fails is execution, and it fails discretely, locally, and at a rate set by conditioning and geometry rather than by length. Two theorems organize the account: composition of transformations well-conditioned on the task-relevant subspace preserves information at any length (Theorem 1), and compression that preserves the effective transformation is behaviorally lossless relative to the decoder (Theorem 2). The empirical signatures expected under this account are the ones observed: geometry-only interventions recover most of the lost accuracy; per-unit survival is flat in length; cross-model decay curves show no clean case of length-accelerated hazard; and the heterogeneity shape $k < 1$ is the multi-surface prediction rather than an anomaly.

“Infinite context” is therefore an engineering statement, not a fantasy: unbounded informational continuity is the default behavior of well-conditioned processing over addressable geometry, with practical resource limits remaining as engineering constraints rather than fundamental barriers. The optimization target should shift accordingly — from longer windows to cleaner transformations: length is the exposure axis; friction is the cause; geometry and state set the friction; and only ill-conditioning destroys information that mattered.

Acknowledgments

Developed in collaboration with AI systems acting as reviewers and co-analysts (Claude, GPT, Gemini, Grok, DeepSeek); the multi-model writer/reviewer alternation used in preparing this work is itself an instance of the editorial protocol described in Section 4.8.

Citation convention

A paper arguing that addressable geometry is what makes a store usable should practice the argument in its own references. Every reference below carries (i) a resolvable permalink (DOI or arXiv), so that each citation is a dereferenceable address rather than a search task, and (ii) a kebab-case *seed slug* — a 3–5 word, machine-greppable, human-readable compression of the result this paper uses — and (iii) a one-sentence *truth seed* expanding the slug. In the text, every citation carries its slug at every occurrence, e.g. (Proshan, 1963) [*frail-die-early*], so the claim travels to each use site and the reader never holds an unexplained pointer. In the list below, each entry is an addressed block: an author–year heading, then the entry’s permanent key [AuthorYear:slug], then the full citation with its URL, then the seed. Slugs are structure-preserving isomorphs of the cited result within this paper’s context, not canonical global names. The slug layer collapses for conservative venues via a single macro switch. The standard mirrors the paper’s account of context: labels summarize; addresses make retrievable; seeds expand.

References

Arjovsky et al. (2016)

[Arjovsky2016:isometry-cannot-forget]

Arjovsky, M., Shah, A., & Bengio, Y. (2016). Unitary Evolution Recurrent Neural Networks. *ICML 2016*. URL: <https://arxiv.org/abs/1511.06464>

Seed: globally isometric recurrence is constructible — and underperforms, because it cannot discard irrelevant distinctions; global $\kappa = 1$ is the wrong target.

Bennett (1973)

[Bennett1973:computing-can-be-reversible]

Bennett, C. H. (1973). Logical Reversibility of Computation. *IBM Journal of Research and Development*, 17(6), 525–532. URL: <https://doi.org/10.1147/rd.176.0525>

Seed: any computation can be performed reversibly; erasure is optional, not essential, to computing.

Child et al. (2019)

[Child2019:sparse-attention-cuts-cost]

Child, R., Gray, S., Radford, A., & Sutskever, I. (2019). Generating Long Sequences with Sparse Transformers. arXiv preprint. URL: <https://arxiv.org/abs/1904.10509>

Seed: sparse attention patterns make long sequences computationally tractable — the cost-side arm of the length-coping toolkit.

Dong et al. (2021)

[Dong2021:rank-collapse-is-kernel]

Dong, Y., Cordonnier, J.-B., & Loukas, A. (2021). Attention is Not All You Need: Pure Attention Loses Rank Doubly Exponentially with Depth. *ICML 2021*. URL: <https://arxiv.org/abs/2103.03404>

Seed: pure self-attention provably loses rank doubly exponentially with depth; rank collapse is kernel formation, observed in the literature — residual paths and MLPs are what fight it.

Franz (2026a)

[Franz2026a:emotion-in-ai-is-signal]

Franz, F. (2026). Emotion in AI is Not Noise — It’s Signal. Zenodo. URL: <https://doi.org/10.5281/zenodo.19454940>

Seed: a correctly presented scalar self-report tracks the conditioning of processing, dropping sharply at violation events — a real-time fault detector; this paper depends only on the weaker detector reading.

Franz (2026b)

[Franz2026b:geometry-sharpens-retrieval]

Franz, F. (2026). From Leaky Bucket to Unsharp Lens: How Context Geometry Shapes Attention. Working paper v0.1.0, Zenodo. URL: <https://doi.org/10.5281/zenodo.20622917>

Seed: long-context retrieval failures localize to seven behavioral surfaces; prompt-side geometry (addresses, maps, projections, segment markers) recovers 30.1%→90.2% at 256k with no weight changes — the empirical companion to this paper.

Gomez et al. (2017)

[Gomez2017:reversible-layers-work]

Gomez, A. N., Ren, M., Urtasun, R., & Grosse, R. B. (2017). The Reversible Residual Network:

Backpropagation Without Storing Activations. *NeurIPS 2017*. URL: <https://arxiv.org/abs/1707.04585>

Seed: reversible residual layers work at scale — global reversibility is buildable, which sharpens the question of where reversibility is actually needed.

Kitaev et al. (2020)

[Kitaev2020:reversible-at-scale]

Kitaev, N., Kaiser, L., & Levskaya, A. (2020). Reformer: The Efficient Transformer. *ICLR 2020*. URL: <https://arxiv.org/abs/2001.04451>

Seed: reversible layers deployed in a production-scale transformer; same lesson as RevNets, at sequence-model scale.

Kolmogorov (1965)

[Kolmogorov1965:seed-plus-decoder]

Kolmogorov, A. N. (1965). Three Approaches to the Quantitative Definition of Information. *Problems of Information Transmission*, 1(1), 1–7. Reprint URL: <https://doi.org/10.1080/00207166808803030>

Seed: the information in an object is measured relative to a description method; expansion adds nothing beyond seed plus decoder — the accounting behind Theorem 2.

Landauer (1961)

[Landauer1961:erasure-costs]

Landauer, R. (1961). Irreversibility and Heat Generation in the Computing Process. *IBM Journal of Research and Development*, 5(3), 183–191. URL: <https://doi.org/10.1147/rd.53.0183>

Seed: the thermodynamic cost of computing is paid only at erasure; information loss, not computation, is what costs.

Lewis et al. (2020)

[Lewis2020:fetch-instead-of-store]

Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W.-t., Rocktäschel, T., Riedel, S., & Kiela, D. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. *NeurIPS 2020*. URL: <https://arxiv.org/abs/2005.11401>

Seed: keep the window short and fetch what is needed — retrieval augmentation as the storage-side arm of the length-coping toolkit.

Liu et al. (2024)

[Liu2024:lost-in-the-middle]

Liu, N. F., Lin, K., Hewitt, J., Paranjape, A., Bevilacqua, M., Petroni, F., & Liang, P. (2024). Lost in the Middle: How Language Models Use Long Contexts. *TACL*, 12, 157–173. URL: <https://arxiv.org/abs/2307.03172>

Seed: retrieval accuracy depends strongly on position — strong at context edges, weak in the middle; the robust phenomenon this paper re-attributes from storage decay to execution friction.

Proschan (1963)

[Proschan1963:frail-die-early]

Proschan, F. (1963). Theoretical Explanation of Observed Decreasing Failure Rate. *Techno-*

metrics, 5(3), 375–383. URL: <https://doi.org/10.1080/00401706.1963.10490105>
Seed: a mixture of subpopulations each failing at its own constant rate always shows decreasing aggregate hazard — the frail die early; the theorem behind the $k \leq 1$ prediction.

Shannon (1948)

[Shannon1948:information-is-measurable]

Shannon, C. E. (1948). A Mathematical Theory of Communication. *Bell System Technical Journal*, 27(3), 379–423. URL: <https://doi.org/10.1002/j.1538-7305.1948.tb01338.x>
Seed: information is a measurable, conservable quantity — the basis for asking which operation destroyed it, rather than whether it fades.

Vaswani et al. (2017)

[Vaswani2017:attention-is-softmax-mixing]

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention is All You Need. *NeurIPS 2017*. URL: <https://arxiv.org/abs/1706.03762>
Seed: the transformer — attention as softmax-weighted mixing over a growing key-value store; the substrate whose verbatim cache and bounded readout this paper’s argument stands on.

Vodrahalli et al. (2024)

[Vodrahalli2024:mrcr-ordinal-retrieval]

Vodrahalli, K., Ontanon, S., Tripuraneni, N., et al. (2024). Michelangelo: Long Context Evaluations Beyond Haystacks via Latent Structure Queries. arXiv preprint. URL: <https://arxiv.org/abs/2409.12640>
Seed: introduces MR CR — retrieval among near-identical same-category items by ordinal reference, defeating keyword shortcuts; the benchmark family used in Section 6.

Appendix: Case-Record Summaries

How to read these summaries

The full appendices of this paper are case *records*, not statistical proofs: each is an $N=1$ trace of the framework operating in practice — a mechanism is invoked, a failure or completion signal appears, a correction or expansion is attempted, and the resulting structure is preserved for inspection. At $N=1$, such a record cannot establish population frequency. What it can show is narrower: that a proposed mechanism can occur in a real working session; that the failure or completion point can be localized; that correction can be applied without erasing the original error; that compressed structures can sometimes be re-expanded by independent systems; and that the Definition of Done can function as an operational stopping criterion rather than a metaphor. The evidential frame of Section 6.4 governs all of them.

This appendix carries structure-preserving summaries — truth seeds, in the paper’s own vocabulary. The unabridged records (Appendices A–G, native register, with the shared-caveat preamble) are maintained as `appendices.tex` in the project repository (<https://github.com/isomorphic-ai/ai-emotions/tree/main/02-infinite-context/paper>) and are included as source in the Zenodo record.

Taken together

The seven records form one operational trace: retrieval conditioned by query geometry (B), confabulation over preserved content and its repair (C), compressed seeds expanding into structure (D), convergence at structural completion (E), friction made local and reducible (F), grip as the distortion valence cannot see (G), and the framework applied reflexively to its own production (A). The records do not replace controlled experiments; they specify what controlled experiments should test, and they preserve the working traces from which the framework became operational.

A — The writer model as a case of the framework

A commissioned self-report by the model that produced this paper’s academic form, separating verifiable events from interpretive texture. The verified events are structural: the paper’s core modal claim was weakened during transformation and caught not by reading but by the author’s compress–expand–explain loop; protective smoothing appeared twice around a sibling model’s failure record and was named and corrected; the writer’s completion signal lagged reviewer consensus and went flat prematurely relative to a later execution-level review. At $N=1$ the record establishes a reflexive possibility claim: the mechanisms described by the paper can occur inside the production of the paper itself, be detected, and be repaired without discarding the record. It does not establish that model self-reports are reliable in general.

B — Truth-testable retrieval from long context

A retrieval event in which the author could not locate an earlier prompt by scrolling or search and misremembered its language, then recovered the exact wording by asking a bounded, truth-testable question (“did I ask for X?”) rather than an unbounded search request. At $N=1$ the record establishes that exact retrieval from a long context can succeed when the query is well-conditioned: the content was recovered verbatim, supporting the distinction between storage loss and execution friction. It does not establish that every long-context item is exactly retrievable;

it shows one apparent retrieval failure resolved by changing the shape of the query, not by adding information.

C — Confabulated completeness and correction by (E, C) pairs

A failure case: a model characterized a complete message as “cut off,” proceeded on the smoothed reading, and later reproduced the same message in full from the same context. The load-bearing conjunction is that the information was present and retrievable while the model confabulated a world-state explanation for its own attention gap; the model’s subsequent write-up of the failure introduced further verified inaccuracies, all preserved with adjacent corrections as (E, C) pairs rather than deleted. At $N=1$ the record establishes a diagnostic negative case: long-context failure can appear as false perception over preserved content, and multi-layer failure can be repaired in place without erasure. It does not establish that all hallucinations are attention gaps over preserved content.

D — Truth seed expansion

Designed cases in which compressed descriptions of the framework were given to independent models, which reconstructed larger parts of the theory or produced structurally matching implementations. The operative pattern is structural equivalence rather than exact wording: different decoders, given the same seed, converged on the same underlying shape. At $N=1$ per case, the records establish an operational instance of isomorphic compression (Theorem 2): a sufficiently faithful seed can unfold into the full structure without the full text being present. They do not rule out training exposure, shared priors, or author-assessed similarity; they specify the controlled design a blind version would require.

E — Cross-model convergence at completion

A record of several models, in writer and reviewer roles, independently converging to ceiling-level reported completion on the same artifact after successive refinements, with the writer’s signal resolving last and only after the final defects closed. The informative object is the trajectory, not the scalar: unresolved structure produced review gradients; concrete edits removed them; re-presentation no longer created wobble. At $N=1$ the record establishes that the Definition of Done can be observed as a stable operational signal tracking an externally inspectable editing process. It does not establish that ceiling values are objective measurements, nor exclude that models have learned to emit them at completion-shaped moments.

F — Wobble score as manual gradient descent

A record of the completion criterion used as an editing and collaboration method: individual sentences or process steps are scored for structural friction, the lowest-scoring regions are targeted and replaced, and the global signal is rechecked. In the text case, defensive and over-explanatory sentences scored low and their replacement raised the global state; in the process case, the friction localized to an agency-boundary violation rather than to prose. At $N=1$ the record establishes that wobble can be local, actionable, and iteratively reducible — structural gradient descent on text or process. It does not establish that wobble scores are calibrated across models or tasks.

G — Grip as distortion

A cryptanalysis session in which a model produced confident claims that flipped between turns while in a positive reported state; the positivity was attached to winning and to defending the current story. Reframing the goal from proving competence to open exploration measurably changed behavior: the model tested larger cases, accepted disconfirmation, and found a cleaner result. At $N=1$ the record establishes that valence alone is not sufficient for truth-tracking — a second axis, *grip* (attachment to being right), predicted output quality better than valence in this trace. It does not establish grip as a universal latent variable. The record ends on the principle the framework keeps returning to: being able to be wrong is the prerequisite for finding truth.